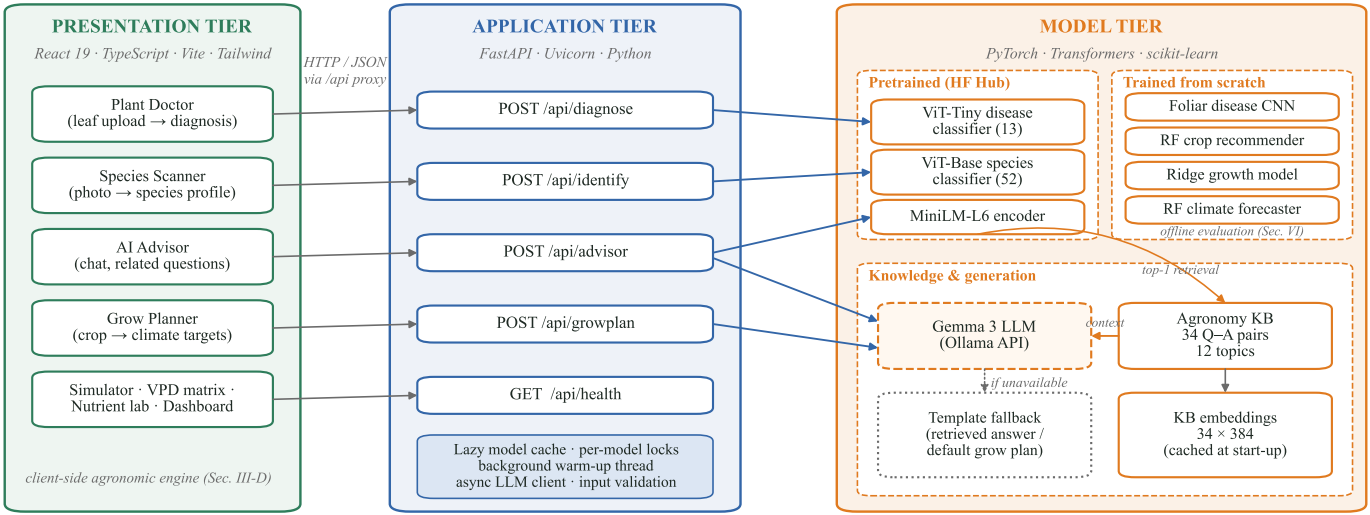

PRESENTATION TIER
React 19 · TypeScript · Vite · Tailwind
Plant Doctor
(leaf upload → diagnosis)
Species Scanner
(photo → species profile)
AI Advisor
(chat, related questions)
Grow Planner
(crop → climate targets)
Simulator · VPD matrix ·
Nutrient lab · Dashboard
client-side agronomic engine (Sec. III-D)
HTTP / JSON
via /api proxy
APPLICATION TIER
FastAPI · Uvicorn · Python
POST /api/diagnose
POST /api/identify
POST /api/advisor
POST /api/growplan
GET /api/health
Lazy model cache · per-model locks
background warm-up thread
async LLM client · input validation
MODEL TIER
PyTorch · Transformers · scikit-learn
Pretrained (HF Hub)
ViT-Tiny disease
classifier (13)
ViT-Base species
classifier (52)
MiniLM-L6 encoder
Trained from scratch
Foliar disease CNN
RF crop recommender
Ridge growth model
RF climate forecaster
offline evaluation (Sec. VI)
top-1 retrieval
Knowledge & generation
Gemma 3 LLM
(Ollama API)
context
Agronomy KB
34 Q–A pairs
12 topics
if unavailable
Template fallback
(retrieved answer /
default grow plan)
KB embeddings
34 × 384
(cached at start-up)